

Hratch Ghanime

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EDUCATION

University of California, Santa Barbara

Bachelor of Science, Data Science and Statistics

Data Science Club

Santa Barbara, California

Expected June 2026

- Data4Good 2nd Place Winner, DataOrbit 2nd Place winner, PALiSaDS Scholar, Project Showcase Finalist.

Relevant Coursework: Object Oriented in Python/Java, Data Structures in Python/C++/Java, Probability & Statistics I/II/III, Data Science Principles with R & SQL, Data Science Concepts and Analysis, Stochastic Processes, Linear Regression Analysis, Design of Experiments, ML

Experience/Projects

LastBite – Full-Stack Food Discount Platform

Isla Vista, California

Founder & Lead Developer

February 2025 - Present

- Architected a data-driven platform using **PostgreSQL, Express.js, and React** to optimize the redistribution of surplus inventory, supporting a 200+ user beta rollout.
- Designed and managed a **relational database schema** to track real-time inventory fluctuations, enabling dynamic pricing models for surplus meals across local restaurants.
- Analyzed platform performance and waste metrics, identifying a 15–20% average reduction in daily surplus waste through data-backed reporting for participating vendors

Gaucho Course Optimizer

Goleta, California

Machine Learning, Python, SQL

January 2026 - Present

- Architected a robust data engine that correlates historical GPA distributions with professor sentiment and course difficulty to streamline academic planning for UCSB students.
- Automated the extraction of 10,000+ data points using **Scrapy and BeautifulSoup**, aggregating professor reviews and official university grade data into a centralized **SQL database**.
- **Developed a sentiment analysis pipeline** that quantified qualitative student feedback, resulting in a 90% accuracy rate in predicting course difficulty and providing a data-driven recommendation engine for academic planning.

RoadMap - AI-Powered Route Optimization Platform

Remote

Python, YOLOv8, PostgreSQL/PostGIS, pgRouting, FastAPI, Redis

October 2025 - Present

- Developed a **YOLOv8 computer vision pipeline** for real-time pothole detection, supporting two-class and single-class models with confidence-based severity classification (≥ 0.7 severe, ≥ 0.4 moderate) and a pluggable detector architecture via Python Protocols
- Engineered a 3-tier **data ingestion pipeline** processing 62k+ road segments with FHWA-derived IRI distributions across 10 road classes and PostGIS-based spatial smoothing (200m radius neighbor averaging)
- Implemented a **Redis**-backed caching system with SHA256-based deterministic keying, 3-tier TTL strategy, and graceful degradation achieving sub-3-second response on repeated queries

SkillShock

Goleta, California

Robotics Engineering, C++ Programming, Sensor & Control Systems

January 2025 - February 2025

- Built a **RAG pipeline** using **LangChain** and **OpenAI** to provide grounded, context-aware responses by indexing extensive libraries of educational materials.
- Engineered a **vector search workflow** utilizing **Pinecone** to perform high-dimensional similarity searches, implementing custom chunking and metadata filtering to improve retrieval latency by 30%.
- Deployed the platform as a featured entry for the Data4Good Datathon 2026, demonstrating the ability to build production-ready AI solutions that minimize model hallucinations. Won 2nd place.

SwiftScreen | DataOrbit

Goleta, California

Machine Learning, Python, Pytorch, API's

January 2024 - February 2024

- Developed an advanced resume screening system using **PyTorch** and **transformer-based models** (BERT/roBERTa) for generating meaningful resume embeddings. While implementing Zero-shot Learning to classify resumes into categories.
- Created and deployed a **Flask-based API**, enabling real-time resume querying and ranking through cosine similarity.
- Earning 2nd place out of 125 teams for outstanding innovation and technical performance.

Hopin

Remote

Software Intern

June 2021 - June 2022

- **Optimized backend services using Ruby on Rails**, focusing on the scalability of high-concurrency web applications to support large-scale virtual events.
- **Conducted rigorous debugging and code refinement** to resolve performance bottlenecks, ensuring high availability and a seamless user experience during peak traffic intervals.